

Machine learning-based digital district heating/cooling with renewable integrations and advanced low-carbon transition

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ABSTRACT

Intermittent power production with hybrid storages, dynamic grids' interactions for synergistic complementation, advanced energy management, optimal design and robust operation are critical approaches to realise smart district energy systems. Inter-city energy migration framework with energy flexibility can improve efficiency and enhance resilience in response to fluctuations in power supply and demand. However, limited studies focused on up-to-date technology advances and artificial intelligence-assisted control for district energy systems. This study comprehensively reviewed district heating and cooling networks with diversified grids' interactions, smart energy management and control strategy through multi-disciplinary approaches. An inter-city transportation-based energy migration framework was proposed for district energy sharing and regional energy balance. Technical feasibility and prospects of machine learning methods on energy planning and optimisation have also been demonstrated in terms of demand prediction, energy dispatch, surrogate model development for uncertainty analysis and optimisation, geometrical and operating parameter design. A district energy network was formulated, involving on-site renewable generations, waste heat recovery from centralized power plants, multi-diversified energy storages, advanced energy conversions for distributed renewable energy sharing. Several technical challenges were identified as avenues for future research, including benchmarks for selection of most suitable energy storages considering intrinsic differences and local conditions (e.g., climate and geographical conditions), energy congestions between renewables and hybrid grids, optimisations with advanced algorithms, and multi-criteria decision-making to promote willingness and readiness for stakeholders' participations.

<i>Nomenclature</i>	FHD	following one of the electricity or cooling demand with higher primary energy consumption
<i>Abbreviation</i>	FLD	following one of the electricity or cooling demand with lower primary energy consumption
<i>BIPVs</i>	Building Integrated Photovoltaics	FCD following the cooling demand method
<i>CFCs</i>	chlorofluorocarbons	FED following electricity load method
<i>COP</i>	Coefficient of performance	FC Fuel cell

(continued on next column)

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<i>CHP</i>	Combined Heat and Power	<i>GSHP</i>	Ground Source Heat Pump
<i>CFCs</i>	chlorofluorocarbons	<i>HCFCs</i>	hydrochlorofluorocarbons
<i>DES</i>	distributed energy systems	<i>ML</i>	Machine Learning
<i>DCS</i>	district cooling systems	<i>NPV</i>	Net Present Value
<i>DHC</i>	District heating and cooling	<i>RTP</i>	real-time price
<i>DC</i>	District cooling	<i>STC</i>	Solar Thermal Collector
<i>DH</i>	District heating	<i>TOU</i>	time-of-use price

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1. Introduction

The decarbonization with renewable and sustainable developments of district energy systems has been regarded as a critical target for the

modern society and natural ecosystems [1,2]. Low-carbon smart cities [3] with advanced simulation tools [4,5] are significant for carbon-neutral transition [6] and energy resilience enhancement [7,8]. Fig. 1 demonstrates the evolution of district energy networks from the

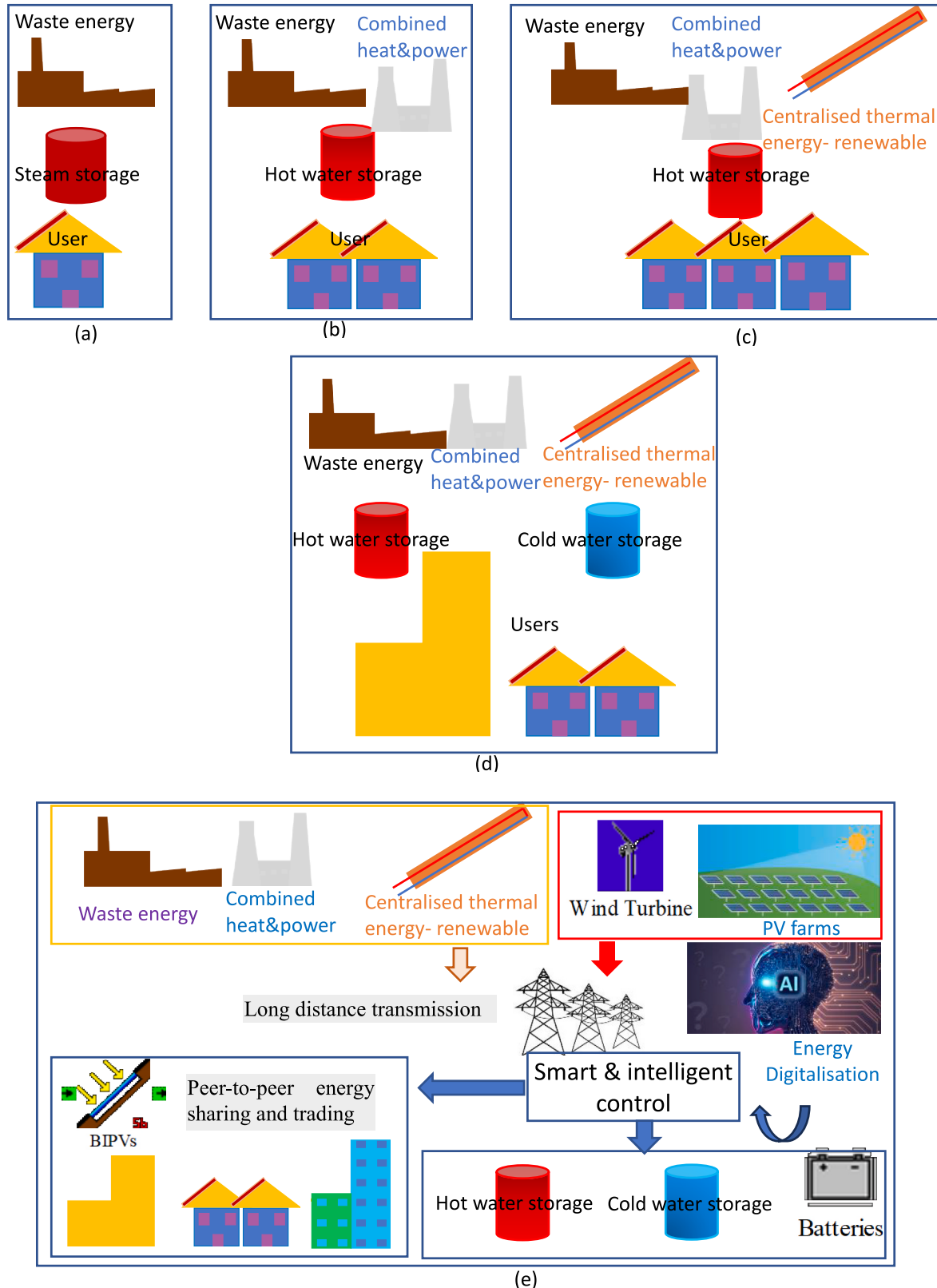


Fig. 1. Demonstration of 5th district energy networks: (a) 1st heat recovery; (b) 2nd district heating; (c) 3rd district heating; (d) 4th district heating and cooling; (e) future digital energy networks.

1st district heating with heat recovery to future digital energy networks. Compared to the 1st district heating, the 2nd district heating integrates the combined heat and power units. Centralised thermal energy for renewable utilisation has been integrated in the 3rd district heating. Furthermore, the 4th district and cooling integrate both cooling and heating storage tanks for renewable penetration improvement. Lastly, the future digital energy network will include a lot of new elements, like energy digitalisation, smart & intelligent control, peer-to-peer energy sharing and trading [9–11], etc. Two concepts related to expanded energy boundaries can be noticed along the evolution of district energy networks from the very early district heating (DH) system to the latest generation, i.e., the increased consideration in waste heat recovery and renewable penetration [1], and more active participation of end-users [1]. District energy systems [12,13] have attracted researchers' interest from perspectives of modelling approaches [14], clean energy production and use [15,16], advanced energy conversions [17,18], high-efficient energy storages [19], and hybrid grids' interactions [20, 21]. Over the past several decades, studies have been mainly on district energy systems in different climates. However, a holistic review is missing in combining DH/district cooling (DC) networks, hybrid renewables' integration, diversified energy storages with multiple energy forms and advanced energy conversions, and hybrid grids' interactions. Furthermore, rational analysis on multi-diversified energy systems and associated optimisation methodology is necessary but rare in academia, especially for smart district energy system, in terms of practical feasibility, multivariable and multi-objective optimisations, and the trade-off between computational complexity and optimisation accuracy. Environmental, Science and Government (ESG) and Sustainable Development Goals (SDGs) require combined efforts from countries/regions, research, industry, and policymakers. Di Silvestre et al. [22] explored how decarbonization, digitalisation and decentralization change the key power infrastructures, involving technological evolution, digital revolution, distribution levels and architectures. Inayat and District cooling system with renewable is highly dependent on local government policy [23]. Zhou [24] comprehensively reviewed carbon neutrality transition with advanced energy policies. Results indicate the global policies played pioneer roles for sustainability transition.

Over the past several decades, the aggravation in the energy shortage crisis and the daily increased energy demand leads to a stricter requirement on energy consumption. In response, different novel energy systems from district levels and techno-economic feasibility in different climates have been proposed, studied and investigated. A state-of-the-art literature review on renewable systems with hybrid energy storage for district heating and cooling (DHC) is critical, and potential challenges need to be outlined to promote the widespread application in the future. Energy storage systems are diversified, including thermal storage systems (sensible, latent, and thermochemical storages) and electricity storage systems (such as electrochemical battery, compressed air energy storage, supercapacitor, fly-wheel and hydrogen system). Advanced energy conversions include solar-to-heating, solar-to-cooling, solar-to-electricity, and heating-to-cooling conversions. The hybrid grids' interactions include DHC grids, power grid, natural gas grid, biofuel grid and hydrogen grid.

The alleviation of fossil fuel depletion, green-house gas (GHG) emission reduction, and energy system stability and efficiency enhancement require renewable penetration, efficient energy storage, and smart grids with optimal design and robust control. In respect to district energy systems, multi-criteria include system efficiency [25], electric power saving [26], primary energy consumption [27], equivalent carbon dioxide emissions [28,29], economic considerations (including the investment cost [30] and operational cost [27]) and life-cycle analysis [28]. The straightforward multivariables related to these objectives involve both design and operating variables. To improve energy performance, researchers have conducted both parametric analysis and optimisation. In respect to the parametric analysis, effective strategies were explored to improve energy performance, such

as the thermal storage in district heating [26], optimal configurations of power plants based on decentralized heating systems [27], and cloud-based demand-response strategies for DH networks [30]. In respect to system optimisation, adopted optimisation algorithms include linear programming [29], genetic algorithms [31], and hybrid mixed-integer linear with a three-stage heuristic approach [32]. One of the critical challenges in optimisation is the computational complexity, which requires the time-consuming computation process with thousands of threads to identify optimal solutions. The technical feasibility of machine learning methods on the enhancement of optimisation has been reviewed. In addition to optimisations, prospects of machine learning techniques have also been demonstrated, in terms of prediction of district energy demand [33] and robust model development [34,35], surrogate model development for uncertainty and optimisation analyses [36], optimal geometrical and operating parameters, and efficient and robust energy management and control strategy [37,38].

Based on the above literature review, research gaps can be noticed.

- 1) Optimal planning, design and operation of multi-energy systems require computational complexity, which requires the time-consuming computation process with thousands of threads to identify optimal solutions;
- 2) Strategies for energy resilience and flexibility enhancement of district heating and cooling energy networks have not been systematically reviewed, in terms of heating networks' design, the demand and grid-responsive control, and the flexible management of hybrid energy storages in accordance with hybrid grids' requirements;
- 3) Roles of machine learning techniques in energy digitalisation have not been characterised. Functions of machine learning in complicated multi-energy systems are unclear.

Research novelty and originality of this study are summarized below.

- 1) An interactive energy network was formulated in this study to provide frontier guidance to promote resilient and smart district energy networks in a future smart city, which systematically integrated on-site renewable generations, waste heat recovery from centralised power plants, multi-diversified energy storage, advanced energy conversions for energy sharing;
- 2) Strategies and effective solutions have been proposed for energy resilience and flexibility enhancement of district heating and cooling energy networks, including heating networks' design, the demand and grid-responsive control, and the flexible management of hybrid energy storages in accordance with hybrid grids' requirements;
- 3) Functions and roles of machine learning in complicated multi-energy systems have been systematically reviewed, in terms of prediction of district energy demand and robust model development, surrogate model development for uncertainty and optimisation analyses, optimal geometrical and operating parameters, and efficient and robust energy management and control strategy.

2. Methodology

A holistic literature review of district energy systems was conducted following the roadmap, as listed in Fig. 2. Both district heating and cooling systems are involved with hybrid renewable integration and hybrid grids' interactions (such as district cooling/heating grids, power grid, natural gas grid, biofuel grid and hydrogen grid). Five grand thermal challenges were identified (e.g., thermochemical/physical thermal storage [39,40], industrial decarbonization, heat pumping systems, long-distance transmission of heat and variable conductance building envelopes), together with potential solutions [12]. Roles of hybrid renewables, novel energy storage systems, and diversified grids' interactions have been analysed. Afterward, smart district energy systems have been reviewed, regarding optimal design, robust operation, energy controls and resilient grids' interactions. Optimizations on

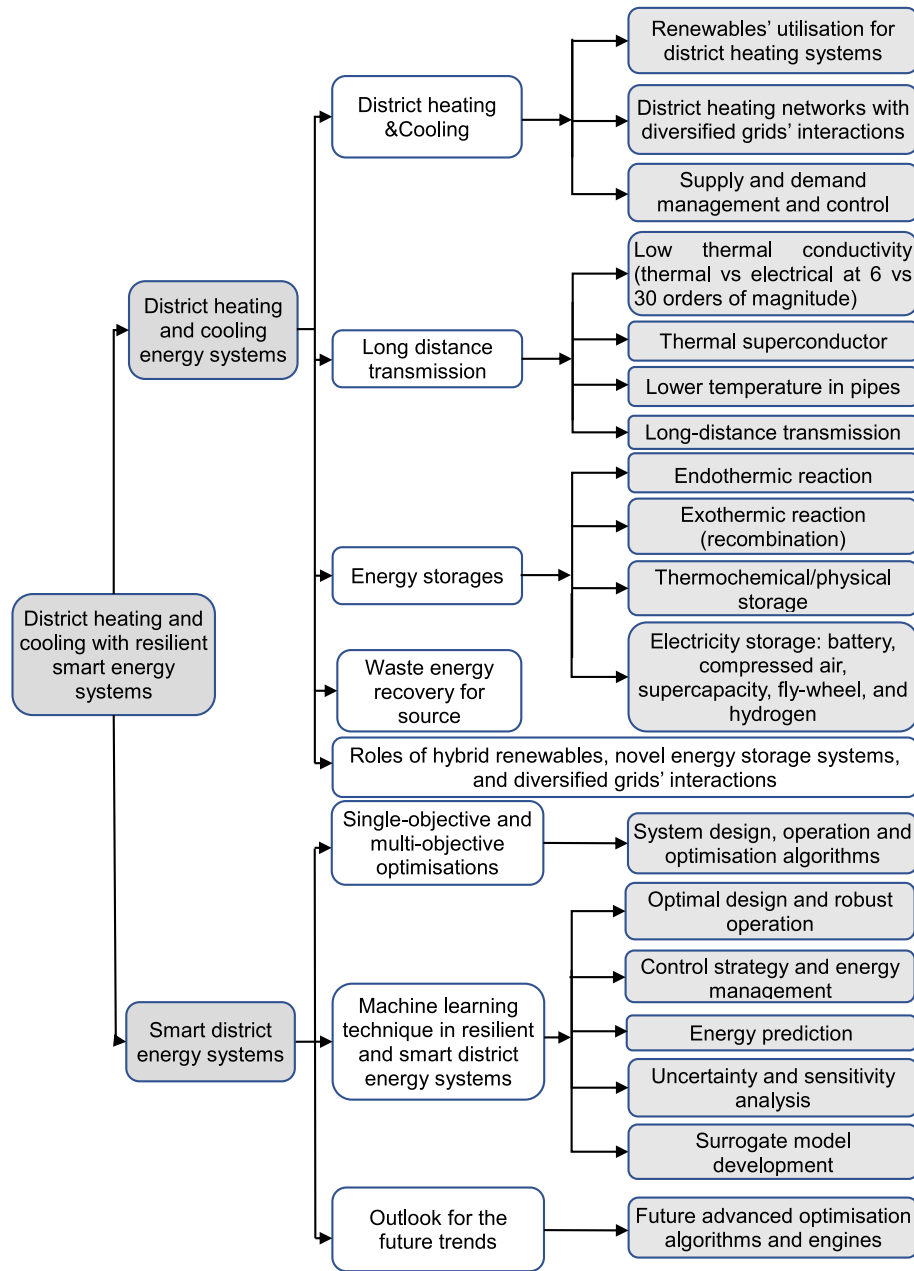


Fig. 2. The roadmap of renewable systems with hybrid energy storages for district heating and cooling.

nonlinear district energy systems were comprehensively presented for the promotion of smart district energy systems. Furthermore, applications of machine learning methods for the realisation towards resilient and smart district energy systems have been analysed, in terms of prediction on district energy demand and robust model development, surrogate model development for uncertainty and optimisation analyses, geometrical and operating parameters' optimal design, efficient and robust energy management and control strategy. Last but not the least, a district energy network has been formulated for the guidance of resilient and smart district energy networks in the future smart city, involving with on-site renewable generations, the waste heat recovery from centralised power plants, multi-diversified energy storages, advanced energy conversions for interactive energy sharing.

In terms of systematic literature selection, the bibliometric approach is adopted by typing keywords in Google scholar. Keywords mainly include district heating and cooling, hybrid energy storage systems, multi-objective optimisations, machine learning, resilient and smart

grids' interactions, and etc. The most impactful articles (e.g., high citation) within latest 5 years are given high priority for the selection. Furthermore, considering the globalisation and diversity of the research, priority is also given to the publications in different continents and climates.

This review includes three sections. The first section systematically reviews district energy systems with renewables and hybrid energy storages, including five subsections, i.e., district heating networks, district cooling networks with energy management and control strategy, energy storages with hybrid grids' interactions, roles of hybrid renewables, novel energy storage systems, and diversified grids' interactions, and an in-depth rational analysis. The second section presents smart district energy systems with energy controls and resilient grids' interactions, including optimisations, artificial intelligence in resilient smart district energy systems, and prospects. In the third section, a highly interactive district energy system was formulated to guide flexible district energy networks in the future smart city, involving on-site

renewable generations, waste heat recovery from centralised power plants, multi-diversified energy storages, advanced energy conversions for energy sharing. Challenges and opportunities are identified from benchmarks to select the most appropriate energy storage. It considers the difference of mechanisms and local conditions (such as climate and geographical conditions), confusions of hybrid grids, multivariable and multi-objective optimisations, and multi-criteria decision-making to promote the willingness and readiness for stakeholders' participations.

3. Systematic literature review of district energy systems with renewables and hybrid energy storages

A holistic overview of district energy systems and research objectives are listed in Table 1. From the perspective of building types, the district energy systems include residential buildings, office buildings, commercial/industrial buildings, university buildings, standalone retail buildings, and building clusters. Energy storages in district energy systems include the sensible, latent, and thermochemical/physical storages for thermal energy storage, the electrochemical battery [41], compressed air energy storage, supercapacitor, fly-wheel, and hydrogen for electricity storage. The involved district microgrids include electrical, thermal, and natural gas grids. District energy management and control can be from demand side management, optimisation-PV power predictions and electricity demand, increase in renewable penetrations with integrated energy storage. Multi-objectives are included in the district energy systems, including system efficiency, electric power saving, primary energy consumption, equivalent CO₂ emissions, economic considerations (initial investment and operational cost), and life-cycle analysis.

3.1. District heating networks, energy management and control strategy

3.1.1. Systematic literature review of renewable energy utilisation for district heating systems

In the context of DH, renewable energy utilisation can be classified into heating sources, system formulation, energy conversions and hybrid

microgrids' interactions. The energy scales cover from single buildings, office building clusters [92], district buildings [14], cities until different regions [93]. Due to the high efficiency and low cost of the solar thermal system, the parabolic collector has been widely studied in adaption to different climates to maximize thermal efficiency. Table 2 lists a holistic overview of district heating systems, including diversified energy sources, energy storages, advanced conversions and hybrid microgrids' interactions. Researchers mainly focused on the novel design of non-tracking concentrating collectors to improve thermal efficiency [94]. Compared to the commercialized non-tracking collector, the new collector can increase the thermal efficiency by 20 %. The solar parabolic-trough collector using rotatable axis tracking shows more competitiveness due to the real-time vertical incidence of solar radiation to the collectors' center [95,96]. Researchers have paid particular attention to the intermittence and instability of solar energy. On the other hand, geothermal energy is more stable, providing almost constant temperatures by the ground with considerable storage capacity. The energy utilisation forms of geothermal energy for DH include tunnel ventilation [73], drilled boreholes-based ground source heat pumps [53, 97], and geothermal assisted air-conditioning systems [93]. In addition to solar and geothermal energies, the combined heat and power (CHP) was regarded as the most reliable energy source for district heating systems. According to different driving sources driving, combined heat and power systems (CHPs) can be classified into the coal-fired CHP plants [98,99], the biomass-driven combined heat and power [100,101] and the waste-to-energy CHP plant [102]. Compared to the coal-fired or natural gas-fired CHP plants, the biomass-driven CHP and the waste incineration CHP plant are more promising. The transportation networks can contribute to the local district heating grids with the implementation of heat recovery. Davies et al. [103] upgraded the recovered heat from the underground railways and contributed to the local district heating network. Along with the implementation of the proposed heat recovery technique, 50 % of carbon savings can be realized. Revesz et al. [104] recovered the heat from underground infrastructure for efficiency enhancement. The research results indicated that the heat extraction rates of ground source heat pumps (GSHPs), which were close to

Table 1
A holistic overview of district energy systems and research objectives.

Building Type	Residential buildings [42, 43]	Office building [44,45]		Commercial [46]/industrial [47]/office building [48]	University building [49]	Standalone retail building [46, 50]	Residential [51] and commercial building clusters [52]	
District Energy Storages	Thermal energy storages Sensible energy storages: ground [53]and molten salt energy storage [51,54, 55]	Latent energy storages: PCMs energy storage [56, 57]	Thermochemical/ physical storage [58,59]	Electric energy storages Battery storage [60]	Supercapacitor [61–63]	Fly-wheel [64]	Compressed air energy storage [65–67]	Hydrogen energy storage [68–70]
Renewable Energy Sources	Solar-to-electricity Solar PV	Solar Tower Molten Salt Photothermal Power Generation [71]		Wind-to-electricity Wind turbine [72]	Geothermal energy Earth-to-air heat exchanger [73]	Ground source heat pump [53]	Biomass Bio-mass driven micro combined heat and power unit [74,75]	
District Microgrids	Electricity Heating/ Cooling Fuel	Electric grid [76,77] Electrical + thermal grid [78] Natural gas grid [77]						
District Energy Management and Control	Demand side management strategies [50] (e.g., adjustment of the cooling [79], demand response [44], grid response [80], Renewable-to-thermal storages [81,82], and occupants' behaviors [14])			PV power and electrical demand [33], model predictive control (MPC) [43]	Renewable penetrations in buildings with extended energy storage boundary: PEVs [83] and HVs [84]	Participation of end-users [85]		
Objectives	Grid interaction [86,87]	Matching capability [45,88]		Equivalent CO ₂ emission [89]	Operational cost [50, 87]	Life cycle cost or net present value [89]	Thermal comfort, power consumption [90]	Energy flexibility [91]

Table 2
The holistic overview of the district heating systems, in respect to diversified energy sources, energy storages, advanced conversions and hybrid microgrids' interactions.

Heating sources	Systems	Energy forms	Storages	Energy conversions	Microgrids' interactions	Scale
Solar thermal energy	Solar thermal collector [105] Solar assisted ground source heat pump [110]	Solar thermal energy	Sensible energy storage [106,107]	Solar-to-thermal [105]	Heating grid [108]	Single buildings, office building clusters [92], district buildings [14], cities, regions [109]
Geothermal energy	Tunnel ventilation [73]; Ground source heat pump [53,97]; Geothermal assisted air-conditioning system [93]; Advanced piping system [116]	Geothermal energy	PCMs latent energy storage [107]	Power-to-thermal [100,101]	Power grid [107], natural gas [111], waste heat [112–114] and hydrogen fuel grid [115]	
Coal/natural gas/incineration	Gas-fired Boiler [117] Coal-fired CHP plants [98,99] Waste-to-energy CHP plant [102]	Coal/electricity/ natural gas/ Biomass		Bioenergy-to-thermal [101, 118]		
Biomass	Biomass co-firing [118] Biomass driven combined heat and power [100,101]					
Nuclear energy	Nuclear Battery [119]; nuclear district heating [120–122]					
Data center Heat recovery	Low-grade waste heat recovery [123–125] Industrial waste heat recovery in district heating [113]; waste-to-energy CHP plant [102]; cascade recovery of flue gas waste heat [114] Engine waste-heat recovery [127,128]		Thermochemical energy storage [1,126]			
	Exhaust heat recovery					
	Heat recovery from transportations					
	Underground railways heat recovery for district heating network [103,104]					

underground railways, could be increased by around 43 %.

As critical and effective breakthroughs for the mismatch between energy generation and energy demand and the load shifting for cost-saving purposes, energy conversion and storage have attracted researchers' interest worldwide. To meet the heating demand, energy conversion techniques include the solar-to-thermal [129,130], power-to-thermal [131], and bioenergy-to-thermal conversions [132, 118]. Thermal energy storages include sensible energy storage [133, 134], latent energy storage with phase change materials [106,107], and thermochemical/physical energy [126]. The systematic integration of advanced conversions and storages in district heating systems can make fast response and sufficient reaction to stochastic demands by avoiding excess energy production, increasing stability of energy networks, and minimising energy congestions.

Regarding the district heating systems, several technical challenges need to be clarified and investigated in the upcoming studies. Firstly, the rotatable axis in the solar parabolic-trough collector for the solar tracking will suffer from performance degradation over thousands of rotating cycles, and the maintenance cost of the rotatable axis needs to be well considered. Secondly, in respect to the solar-assisted or geothermal-assisted GSHPs, technical challenges remain for the high initial cost of drilled boreholes and the unbalance of the annual heat rejection and extraction in the ground. Novel strategies are necessary for the energy balance in different climates. Thirdly, capacity expansion of microgrids and the development of standardized auxiliary facilities will increase the economic investment during the facility upgrade.

3.1.2. District heating networks with diversified grids' interactions

Synergistic complementarity with combined operation on each integrated system, is required to form a smart energy network with hybrid grids' interactions. The sophisticated building energy systems can acquire the ability to manage renewable generations and energy demands, regarding climate conditions, occupants' demands and local grids' requirements. By dynamically managing renewables, demands, storage [135], and grids' interactions, a smart energy system can cover energy demand with high energy flexibility [136], low CO₂ emission, and reduced reliance on grids. An interactive districting system with renewable, storage, diversified grids' interactions has been formulated, as shown in Fig. 3. Thermal energy sources include solar thermal collectors, geothermal energy and recovered thermal energy from plants and industries. Electricity sources include gasoline-based diesel engines, building integrated photovoltaics (BIPVs), and wind turbine. As a promising hybrid energy generation device, the CHP can provide multilevel supply energy for cascade utilisation. Energy sources driving the CHP include bio-fuel, biomass, and waste incineration. The surplus electricity can be stored in hybrid electricity storages, together with heating energy recovery for domestic usage, such as heat recovery from the fuel cell-based hydrogen storage [137] and friction heat recovery from the compressor. The surplus or recovered thermal energy can be stored in daily thermal storage tank or seasonal thermal storage ground [138]. Hybrid grids dynamically balance energy in the formulated energy system via power grid, district heating grid, biofuel grid, natural gas grid, and hydrogen fuel grid [139].

3.1.3. District heating networks, energy management and control strategy

In order to improve the resilience and flexibility of DH networks, technical solutions can be focused on the design of heating networks, demand-responsive control, and management of energy storages under hybrid grids requirements. In addition to the main participants of building sectors, participants of transportation sectors can perform well in the whole system. The main principle is that, mobile thermal storage systems with vehicles can realise the energy sharing and establish the interactive energy network between diversified heating sources and end-users [138]. For instance, as shown in Fig. 4, trucks can carry goods to the factory. It can also take thermal energy, with a thermal energy storage with high energy density, from the exhaust heat in factory back

to cover the heating demand of end-users. In such a scenario, the exhaust heating energy exported to the heating grid from the factory can be reduced and the heating energy imported from heating grid can also be reduced to cover the end-user's heating demand.

From the perspective of heating networks' topology design, the branched pipe network and the annular pipe network can be noticed. In respect to the district heating network with annular pipes, as shown in Fig. 4(b), mobile vehicles with energy storages can systematically connect different buildings, centralized energy storage, exhaust heat sources (like factories), and energy generators (like CHP plants). In addition to mobile vehicles, centralized energy storage systems can contribute to the energy sharing between energy generation and demand units when implementing annular pipe networks. Compared to the sophisticated piping system in branched pipe networks, as shown in Fig. 4(a), energy sharing between different energy generation and demand units becomes more straightforward in annular pipe networks. Along with the participation of mobility and thermal storages, the district heating network with annular pipes, integrating peer-to-peer energy sharing and trading [141], vehicle-to-building interaction, and so on, can be more reliable and flexible than the branched pipe network in terms of power supply reliability, system resilience under extreme and unexpected disruption, and mitigation on grid reliance.

For efficient energy management in the building energy system, researchers have studied demand-responsive control, grid-responsive control strategy, and management of hybrid energy storage. In respect to demand-responsive control, Mishra et al. [142] conducted field testing to study the impact of inlet water temperature-based demand response on indoor thermal comfort. Research results indicate that the demand response will not affect the occupants' perception of indoor thermal environments, while it can provide significant avenues of energy flexibility through thermal mass in the building envelope. Guelpa et al. [143] investigated the demand side management for the construction of optimal control of DH networks, from perspectives of time reschedule of heating systems and modification of parameter settings. They conclude that the peak reduction can be 35 % on the peak request. Guelpa et al. [144] studied demand-side control strategy with different return temperatures (DRT). According to the results, using the DRT regulation strategy can reduce building thermal peak demand by 15 % and peak power by 24 %. In respect to the grid-responsive control strategy, Wang et al. [145] provided a systematic literature review on the development of grid-responsive buildings with fast demand responses to smart grids. Zhou et al. [146] developed an advanced grid-responsive control to provide techno-economic viability and improve energy flexibility to power grid. In respect to the flexible

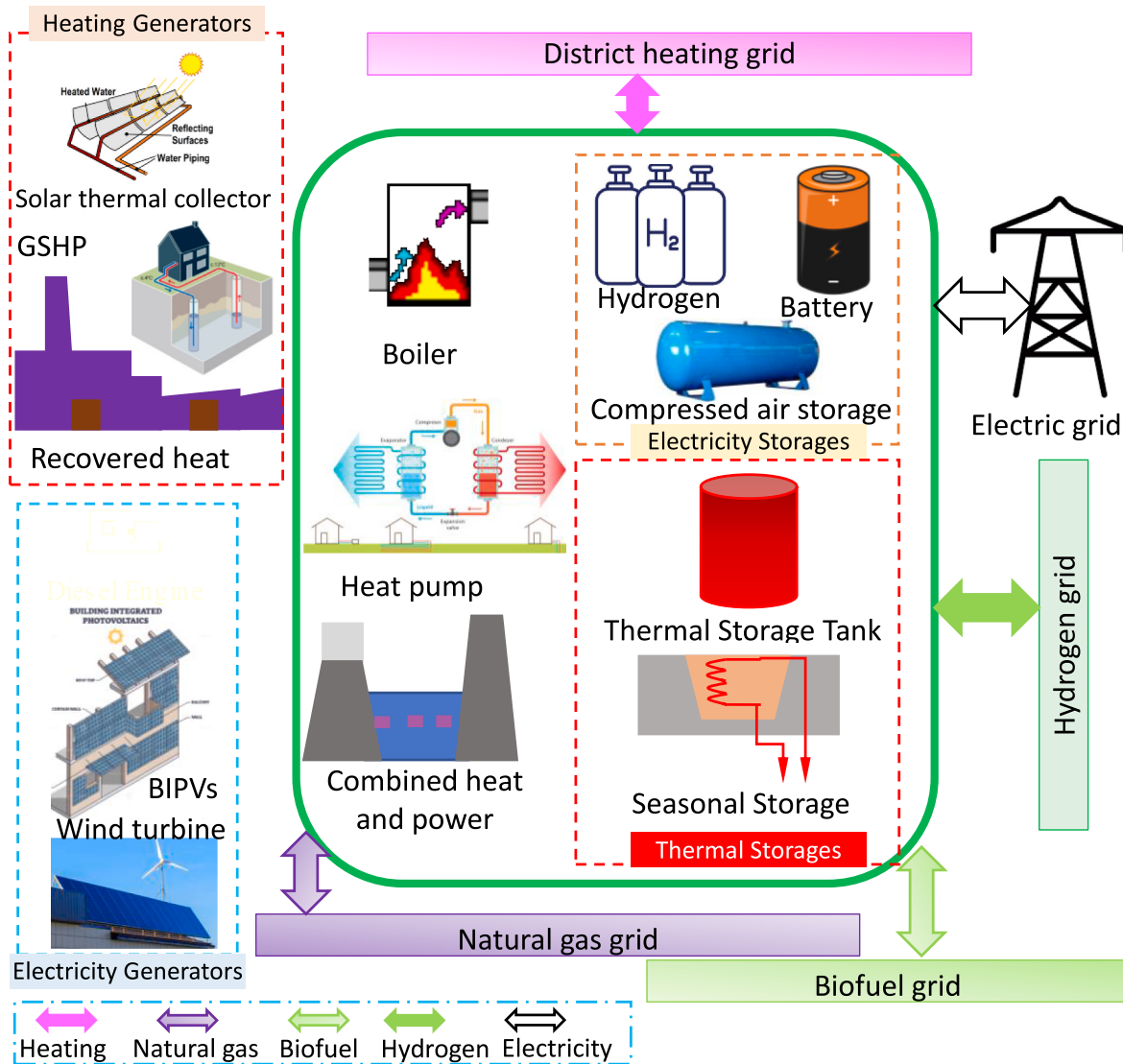
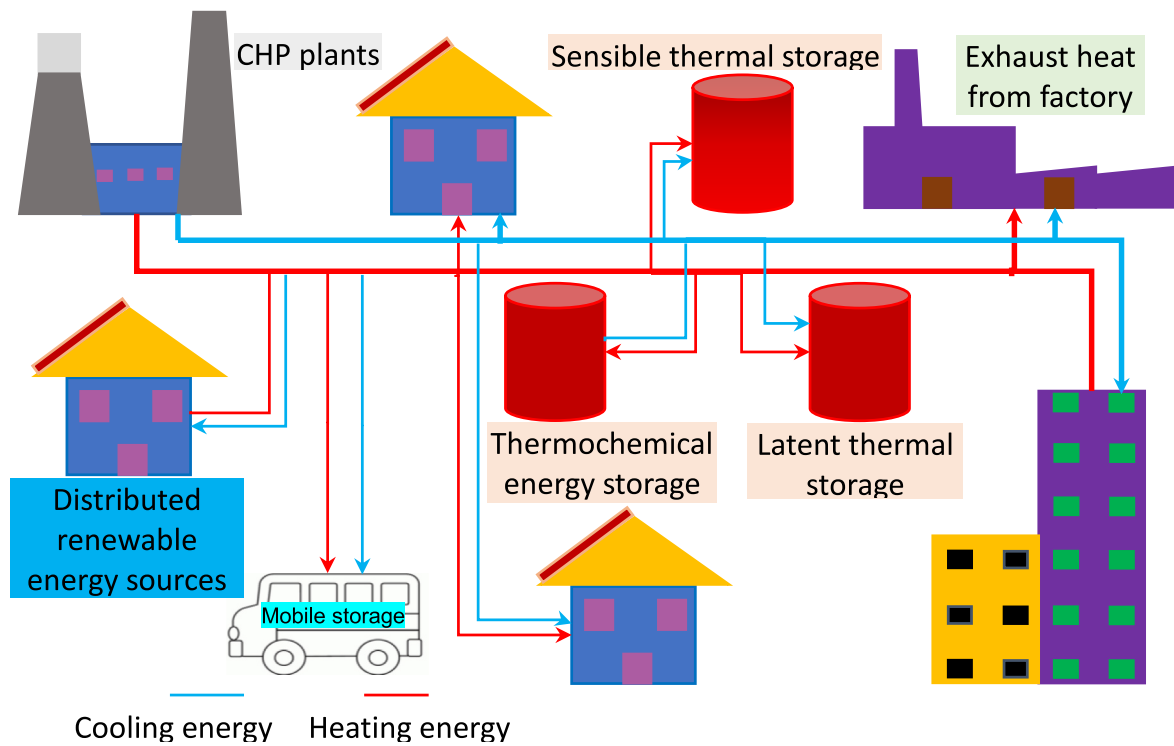
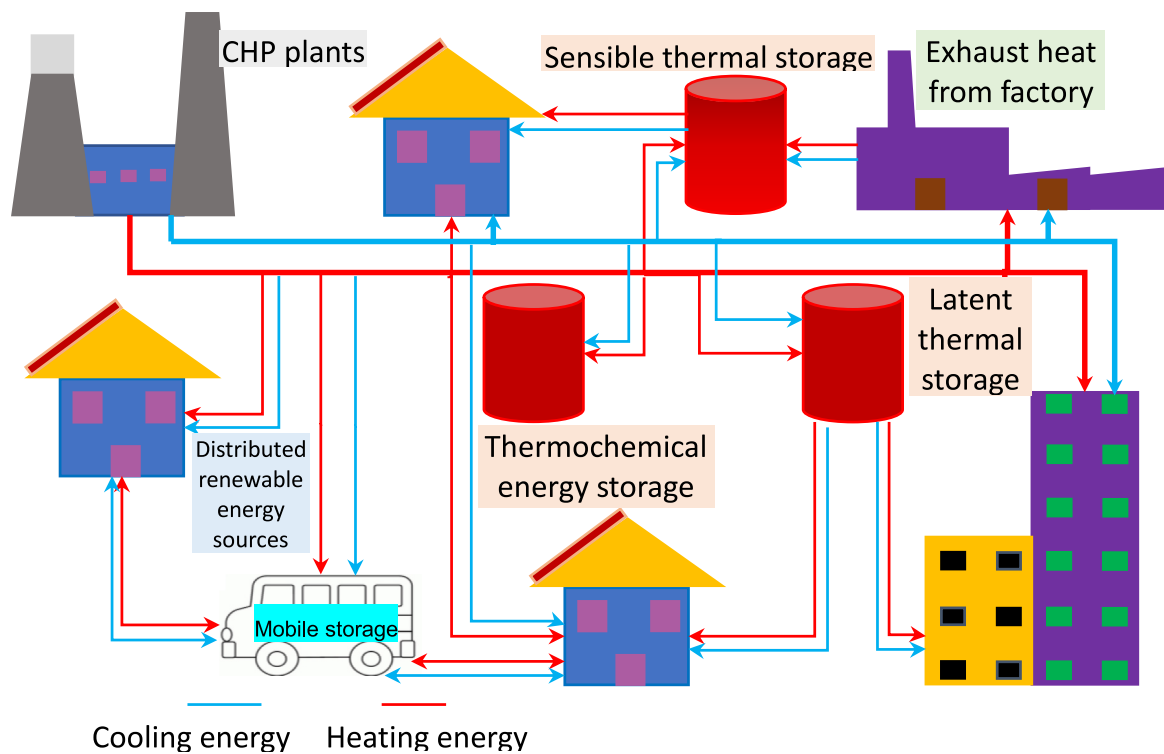


Fig. 3. Renewable-to-electric and thermal storages with multiple energy grids' interactions.. (Note: GSHP refers to ground source heat pump.)



(a)



(b)

Fig. 4. TES connected to district heating network, in respect to (a) branched pipe networks [140]; (b) annular pipe networks [140].

Table 3

Cooling sources	Systems	Storages	Energy conversions	Microgrids' interactions	Scale
Solar thermal energy	Absorption cooling system [160,105]	Sensible energy storage [163,175]	Solar-to-power [151,163]	Smart heating grid [176,177]	Single buildings [149,150], residential building clusters [151], district buildings [152,153], cities, regions [154]
	Adsorption cooling system [160,105]				
Geothermal energy	Desiccant cooling system [178,179]	PCMs latent energy storage [180]	Solar-to-Power&cooling [161,162]	Heating and power grids [78]	
	NH3–H2O refrigeration [164]				
Solar photovoltaics	Single-stage Li–Br water absorption cycle [165, 166]	Electric storage [163,153]	Power-to-thermal [175, 181]	Power grid	
	Absorption heat pump [167]				
Sea water	Geothermal assisted air-conditioning system [169]	Thermochemical energy storage [126]	Heating-to-cooling [105, 181]	Heating and fuel grids [182]	
	Electricity-driven compression chillers [149]				
Biomass	PV assisted cooling system, solar cooling in envelope [161,162], solar PV air-conditioner, Solar thermoelectric cooling-heating system [163]	Thermochemical energy storage [126]	Bioenergy-to-thermal [181]	Heating and fuel grids [182]	
	seawater shower cooling towers [172]				
Coal/Diesel/natural gas	seawater air conditioning [173,174]	Thermochemical energy storage [126]	Bioenergy-to-thermal [181]	Heating and fuel grids [182]	
	Biomass-driven absorption chiller [153]				
Heat recovery	municipal solid waste driven power and cooling [170,154]	Thermochemical energy storage [126]	Bioenergy-to-thermal [181]	Heating and fuel grids [182]	
	biomass boiler integrated absorption chiller [175]				
Exhaust heat recovery	Centralized [181]	Thermochemical energy storage [126]	Bioenergy-to-thermal [181]	Heating and fuel grids [182]	
	and decentralized CHPs [151,170]				
Brake heat recovery	Adsorption cooling with waste heat recovery from internal combustion engine [171]	Thermochemical energy storage [126]	Bioenergy-to-thermal [181]	Heating and fuel grids [182]	
	Regenerative braking				

3.2. District cooling networks, energy management and control strategy

3.2.1. Systematic literature review of renewables' utilisation for district cooling systems

Energy conversions and thermal energy storages are effective to improve energy performance of renewable supported district cooling

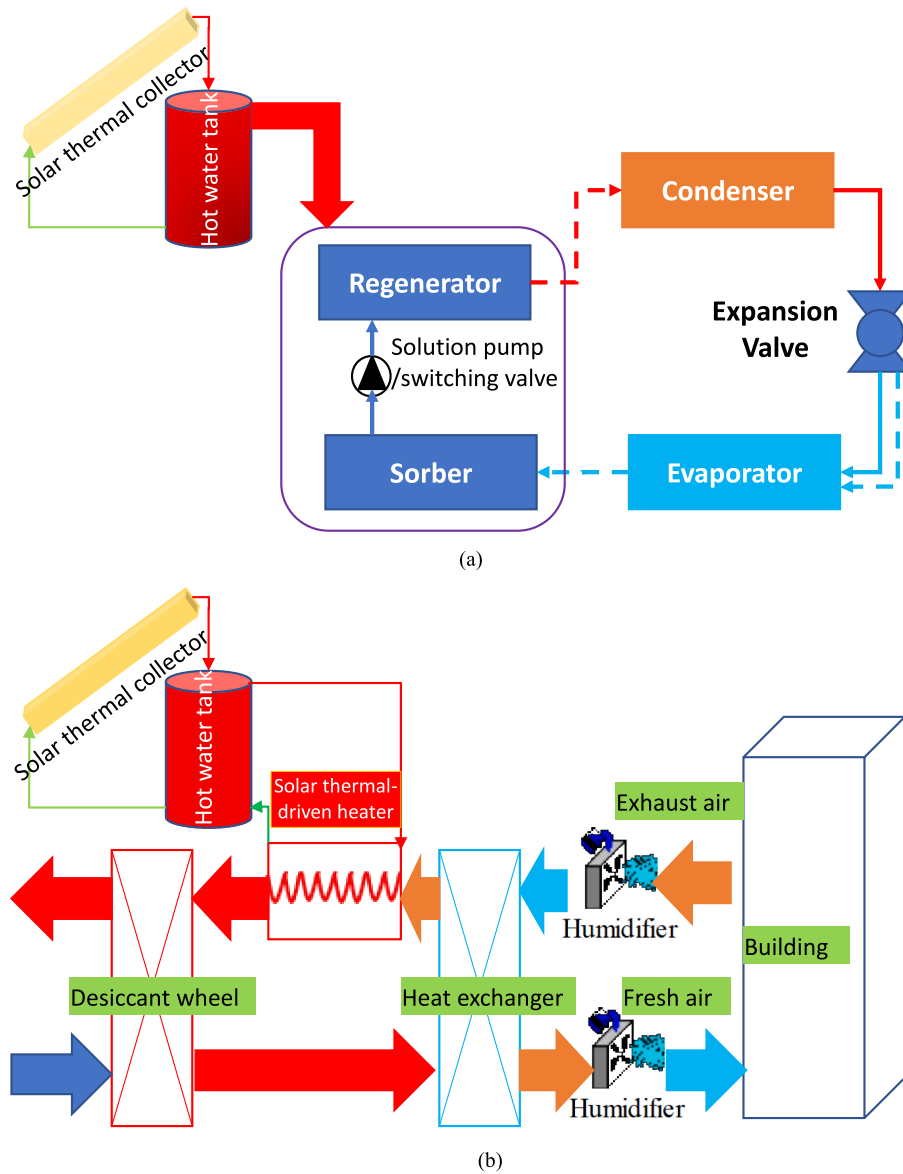


Fig. 5. The mechanism of solar thermal-driven cooling systems: (a) sorption system; (b) desiccant cooling system.

system. The adopted energy conversion techniques include solar-to-power [163], solar-to-power and cooling [161,162], power-to-thermal [175,183], heating-to-cooling [105,181] and bioenergy-to-thermal [181]. Energy storage have been widely applied to enhance the renewable penetration and demand coverage concerning the use of surplus energy from renewables and energy conversion devices. Energy storages for district cooling can be classified into sensible energy storage [163,175], PCMs latent energy storage [180], thermochemical/physical energy storage [126] and electric storage [163,153]. Effective management in DC can make fast response and reaction to cooling demands with an enhancement of stability and resilience of energy networks.

Regarding the DH systems, several technical challenges need to be clarified and investigated in upcoming studies. Firstly, the driving temperature for conventional adsorption cooling systems is often higher than 80 °C [184], requiring research efforts on new sorbent materials to lower the regeneration temperature [185,186]. Secondly, in solar-to-power-to-cool system, the heat transfer fluid, CFCs, in the refrigeration cycle process will result in depletion of the ozone layer and global warming. Although adopting an absorption or adsorption cooling system to replace the vapor compressor can avoid the use of freon, the low efficiency of the sorption cooling systems is the main bottleneck for

its widespread application. Thirdly, there might be less resources (e.g., initial investment cost due to limited budget and limited space) for the district cooling grid from the district multi-energy level, especially with intensified congestion among multi-diversified energy grids. The activations of synergistic functions (such as the complementary battery and hydrogen energy systems under idling mode [187,188]), considering differences of hybrid grids' requirement with diversified energy conversions, are practical solutions to improve the resilience and flexibility of multi-energy systems.

3.2.2. District cooling networks with diversified grids' interactions

Regarding the cooling energy sources for DC, solar-driven cooling systems account for a significant share in all renewable-driven cooling systems. The solar resource is abundant in cooling-dominated regions, such as a subtropical region, Hong Kong. Generally, solar-driven cooling systems include solar-to-power-to-cool systems and solar-to-thermal-to-cool systems. The solar-to-thermal-to-cool system is more environmental-friendly, more resilient to weather changes and potentially less expensive.

As shown in Fig. 5(a), the main principle for the replacement of the compressor is described below. Solar thermal energy is the main driving

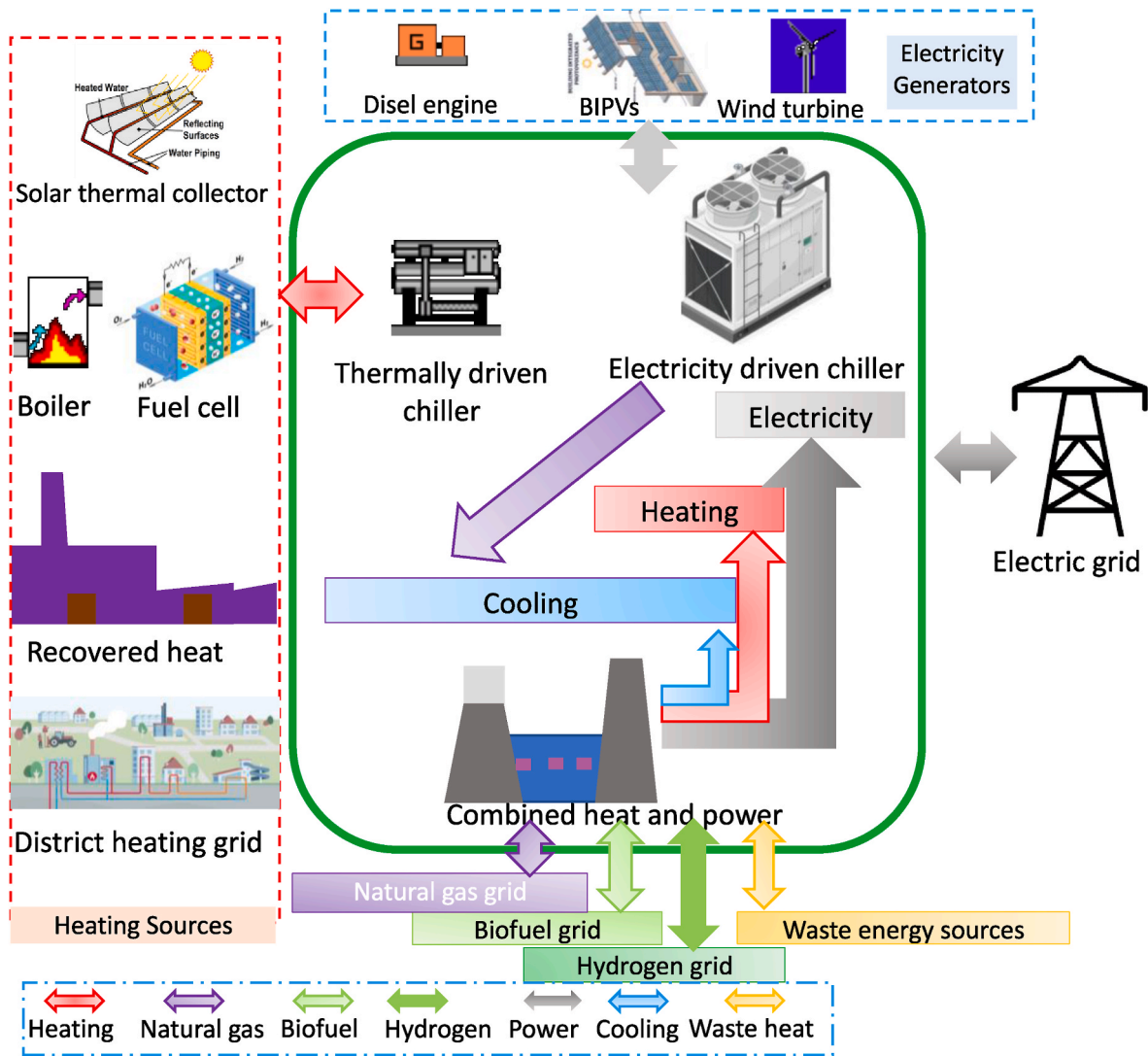


Fig. 6. District cooling networks with diversified grids' interactions.

force to separate the refrigerants from the dilute solution or the adsorbent. After being released, the refrigerants will enter into the condenser, and the concentrated solution or the adsorbent is ready to drive refrigerant and create cooling effect in a new sorption cycle. Fig. 5(b) shows the principle of desiccant cooling. The hot and humid air enters and is thereafter dehumidified by a desiccant component (coated heat exchanger, wheel, etc.) Meanwhile, the air is heated by the adsorption reaction. Afterward, the supply air goes through a heat exchanger for

cooling energy recovery from a cold source (a cooling tower, an evaporator, the exhaust air, etc.) The supply air was cooled and humidified before flowing into the room, if evaporative cooling is used. The humidifier cools the exhaust air from the room and transfers cold energy to the supply air in the heat exchanger. Afterward, the air is reheated by a solar heater, and then transfers heat to the inlet air in the desiccant component. The primary function of solar energy is to heat the exhaust air, which is operated to regenerate the desiccant component and pre-

Table 4

A holistic overview of diversified thermal energy storages.

Thermal storage forms	Types	Applications	Advantages	Disadvantages	Technical challenges
Sensible energy storage [133]	Water and aquifer Rock and bricks	Water storage tank; Ground source heat pump Thermal mass storage	High specific heat capacity and small volume for energy storage Higher temperature endurance; no leakage problem; good conductivity	Upper temperature limit; liquid leakage problem low specific heat capacity and large volume for energy storage	The low energy density results in large storage volume
Latent energy storage	Organic Inorganic	Building envelopes, centralised and decentralised thermal storage and so on	Non-corrosives; no subcooling; Chemically and thermally stable; Adjustable transition zone High enthalpy; High thermal conductivity; Nonflammable; cheap	Low enthalpy and thermal conductivity; Flammability; Low density Corrosion; Phase segregation; Subcooling; Thermal stability	Phase segregation and thermochemical instability over charging/discharging cycles
Thermochemical/physical energy storage	Direct and indirect systems	Concentrating solar power	1) higher energy density; 2) negligible storage loss and higher efficiency for integration with solar power systems.	sophisticated system control, service life, and material compatibility issues and chemical reactions	Performance degradation and material compatibility issues

heat the inlet air.

DC networks have been formulated, as shown in Fig. 6, in respect to thermal and electric generations and hybrid grids' interactions. Thermal energy sources include solar thermal collectors, geothermal energy, and recovered thermal energy from fuel cell systems and centralized power plants. The thermal energy is to drive the sorption chiller to cover cooling loads in district networks. Electricity generation from the gasoline-based diesel engine, BIPVs, and wind turbine can drive the compressor-based chiller to cover cooling load. Furthermore, the CHP can provide multilevel energy for cascade utilisation, with sources from natural gas, biofuel, hydrogen, and waste energy sources. The main function of hybrid grids is to balance diversified energies in the formulated energy system dynamically. Several critical research objectives are worthy of being well investigated: 1) the flexible operations of different devices with diversified energy sources, to improve the overall performance [187] and energy resilience to extreme and unexpected disruptions [189]; 2) synergies within hybrid grids for the flexibility enhancement of DC networks.

3.2.3. DC networks, energy management and control strategy

To enhance the resilience and flexibility of district cooling networks, researchers have mainly focused on demand-side management, integration of renewable and heat recovery systems, and flexible energy management and control strategy. Peltokorpi et al. [190] proposed an organizational system for economic implementation of demand-side control in DC, systematically integrating buildings, systems and customer's understanding.

Sameti et al. [191] indicated that more than 67 % of CO₂ emission reduction could be realized through the solar and heat-driven district cooling, together with cost savings, by optimally designing solar-driven DC systems. Integrating distributed renewable energy with DC can save costs and mitigate grid dependence [192]. Advanced control strategies can cover 81.6 % and 82.9 % cooling loads. Hsu et al. [193] utilised waste heat to drive DC and electricity consumption and CO₂ emission can be reduced by 955.8 MWh and 675 tons per year, respectively.

Cox et al. [194] developed a real-time control to reduce operational costs. Compared to the system with a simple scheduled operation, the real time optimal control with model predictive control strategy can contribute to 16 % of average savings for price.

Challenges should be addressed in respect to renewable integrated DC networks with efficient energy management and advanced control strategies. Firstly, comprehensive considerations are needed for demand-side management within building clusters, including diversity of occupants' behaviors, building types, specific requirements of each grid (e.g., power restraint, ramp rate), conflicts between grids, and peak power mitigation strategies. Secondly, flexible energy management and control have to consider not only the requirements of surrounding energy networks but also barriers and motivation of involved stakeholders, i.e., end-users, building owners, facility managers, aggregators, district and transmission operators.

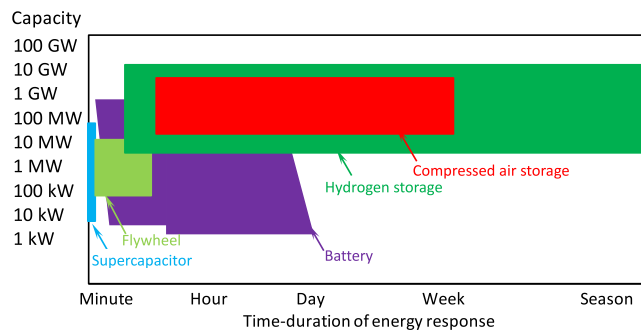


Fig. 7. The storage capacity and time-duration of energy response, with respect to different electric energy storage systems.

3.3. Hybrid energy storages in district energy networks

Thermal storages are necessary to improve the power supply reliability of the energy supply network, the readiness to intermittent thermal demand, and the flexibility through demand-side management in response to requirements of DH grids with multi-diversified sources. Table 4 lists a holistic overview of diversified thermal energy storage. Guelpa et al. [140] reviewed thermal storages in DH, from the pros and cons of thermal storages, such as thermal energy storage forms, key performances indicators, and multi-energy systems. The heating sources include CHP plants, exhaust heat recovery from industries, and distributed renewable energy from on-site buildings integrated renewable systems. Latent thermal storage such as PCMs have been intensively investigated in various applications for the fair storage capacity and good thermos-chemical stability, such as solar thermal storage [195, 196], geothermal energy storage, exhaust heat recovery [197]. Examples also include thermal energy management with electricity-driven devices, such as heat pumps and CHPs. Latent energy storage systems are widely used in distributed energy systems. The thermochemical/physical storage works with the principle to store energy in chemical form. Compared to latent or sensible heat storage, advantages of thermochemical/physical storage can be briefly summarized: 1) higher energy density due to the endo/exothermic reaction; 2) negligible storage loss and higher efficiency. However, technical challenges resulting from sophisticated system control, service life, material compatibility issues need further studies.

In addition to thermal energy storage, electric energy storage is also critical, due to the high energy quality of electricity. Fig. 7 shows a summary of diversified electric storage systems regarding storage capacity and time-duration of energy response. Electricity energy storage systems mainly include the electrochemical battery, compressed air energy storage, supercapacitor, fly-wheel and hydrogen, energy storage. Compared to the minute-based supercapacitor, the battery storage system shows a broader range of time duration for energy response and a larger storage capacity. Hydrogen storage offers the most extensive range of the time-duration in energy response with an elastic storage capacity. Because water is the only byproduct, hydrogen storage is promising in the DH system in terms of clean power production and climate change mitigation. Table 5 lists a holistic overview of diversified electric energy storage, together with advantages and disadvantages for each system. Several technical challenges should be investigated, including dynamic performance depreciation over long-time operations, operational security, enhancement of energy storage density, and the energy loss during the charging process.

The mobility consumption from transportations can be integrated into the smart district energy system, as the transportation sector consumes 28 % of total energy consumption with 23 % carbon emission. In addition to thermal and electric energy storage systems, the regenerative brake system is a reliable and promising energy source. Fig. 8 shows the cascade energy efficiency from transportations and the hierarchical topography for energy utilisation. The input energy in the transportation system includes the energy to the engine and the engine loss. For the engine system, a proportion of energy is sent to the braking system, and the rest becomes the rolling and aerodynamic loss. For braking, the first generation for heat management is burning through resistance which is the mainstream, and the second generation is via battery but the grids often reject the feed-in due to the instability. The third generation is supercapacitor. The energy recovery from the braking system, called energy recuperation, is one of the important sources of input energy. Depending on transportation scenarios, different proportions of recovered thermal and electrical energy can be noticed, as shown in Fig. 8(b) and (c), respectively.

Table 5

A holistic overview of diversified electric energy storages.

Electricity storage forms	Types	Applications	Advantages	Disadvantages	Technical challenges
Electrochemical batteries	Flow battery	Load leveling, power quality and emergency standby	Long life cycle, high power, fast working response	Large space, high cost	1) The battery depreciation over thousands of charging/discharging cycles; 2) The environmental pollution from exhausted retired batteries; 3) The security concerns during operational process.
	Sodium-sulfur battery	Load leveling, peak shaving, power quality and renewable energy integration and management	High energy densities power and efficiency	High cost, safety problems	
	Nickel–cadmium battery	Remote control, lamps, aircraft, and diesel engine starters	Low cost	Short life cycle, low capacity	
	Lead-acid battery	Grid storage, multiple deep-cycle lead–acid batteries	Low cost	High self-discharge rate, short life cycle, slow to charge.	
	Lithium-ion battery	Grid storage, regulation and power management services	Long life cycle, high efficiency and energy densities,	Special charging technology, high cost	
Compressed air energy storage	Constant-volume storage	Tank with high-strength carbon-fiber		Pressure level below a certain safety limit	Energy loss in the compressor
	Constant-pressure storage	Storage vessel on the bottom of the chosen water reservoir	Full exploitation of air without safety concerns from Pressure; Flexible locations for geographic positions, such as coastal lines, floating platforms [198];	High cost	
Supercapacitor	Double-layer capacitors	Consumer electronics; Grid power buffer; Low-power equipment power buffer; Voltage stabilizer; Ultra-Battery	Unlimited cycle life; fast charge/discharge cycles for emergency; no full-charge circuit required; Low impedance; Cost effectiveness	Low energy density around 1/5–1/10 of a battery; Low voltage cells; voltage balancing element required; High self-discharge as compared to electrochemical batteries	Energy density enhancement for storage
	Pseudocapacitors				
Hybrid capacitors					
Fly-wheel	High Velocity Flywheel (angular velocity at 30000–60000 rpm)	Electricity grids for protection against interruptions; wind turbines; motor driven generator to store energy	independent on temperature changes; environmental-friendly with inert or benign materials; lifespan more than 200 years [199]; sealed devices with minimal maintenance; Less thermal loss; More round-trip efficiency; Ease of operation; More peak-load capability	Strong containment vessels for safety precaution	Decrease of energy loss in the friction
	Low Velocity Flywheel (angular velocity up to 10000 rpm)			frictions result in 20 %–50 % of energy loss in 2 h	
Hydrogen energy storage	Low-pressure hydrogen tank	Coverage of electrical and heating demands in district heating networks; transport	Clean energy source, non-toxic	Expensive; safety issue; lower density than gasoline; difficulty in moving with large quantities	Energy density enhancement for storage; Utilisation security
	High-pressure hydrogen tank				

3.4. Roles of hybrid renewables, novel energy storage systems, and diversified grids' interactions

Regarding the solar-to-power conversion, the solar tower molten salt photothermal power generation system shows promising prospects in solar-abundant regions with centralized solar collectors. Fig. 9(a) demonstrates the general principle of the solar tower molten salt photothermal power generation system. Compared to photovoltaic systems, the main advantages of the solar tower molten salt photothermal power generation system can be briefly summarized as 1) less dependence on intermittent solar energy with heat storage of molten salt, which makes the 24-h solar-to-electricity conversion possible; 2) the initial and operational costs fall rapidly, due to fast-developed technology and large-scale batch production; 3) the cogenerated heating energy can be more easily recovered, due to relatively mature technologies in heat transfer enhancement. However, the main challenge for the technology is the low efficiency of the thermally-driven electricity generation process due to the energy quality difference between thermal and electrical energy. Fig. 9(b) shows a renewable-fuel cell energy storage with hybrid grids' interactions. The centralised renewable systems can cover the energy demands of building communities and convert the surplus to clean hydrogen. The byproduct, oxygen, can be used to support the life or as fuel for heating energy generation. The cogenerated heat in the

proton exchange membrane fuel cell can be recovered to cover the energy demand of the building community. The main advantages for hydrogen storage in district energy applications include 1) the clean energy storage with low environmental pollution; 2) the considerable heat recovery is fuel-saving and reduces emissions for DH applications.

In the district energy system, the main function of diversified grids' interactions is to dynamically balance the hybrid renewables and energy demands. Askeland et al. [202] investigated the impact of DH on the flexibility potential of hydropower resources in Norway. The research results indicate that, by shifting from individual electric heating to DH, the maximum load on hydropower facilities can be decreased, and the hydropower system can dynamically balance the year-round electricity demand. Furthermore, the grids' resilience to extreme and unexpected disruption, and fluctuations in the energy supply, can be improved through advanced energy conversions between different energy grids in district energy systems. In respect to energy management, both demand-side management and energy storage can be operated to manage renewable energy, according to climate conditions, users' needs and grids' requirements. The availability of advanced energy conversions can promote renewable penetration, energy sharing, and grid interdependence, with complementarity between diversified energy forms. The complementary interaction between energy efficiency and demand response can bring economic benefits.

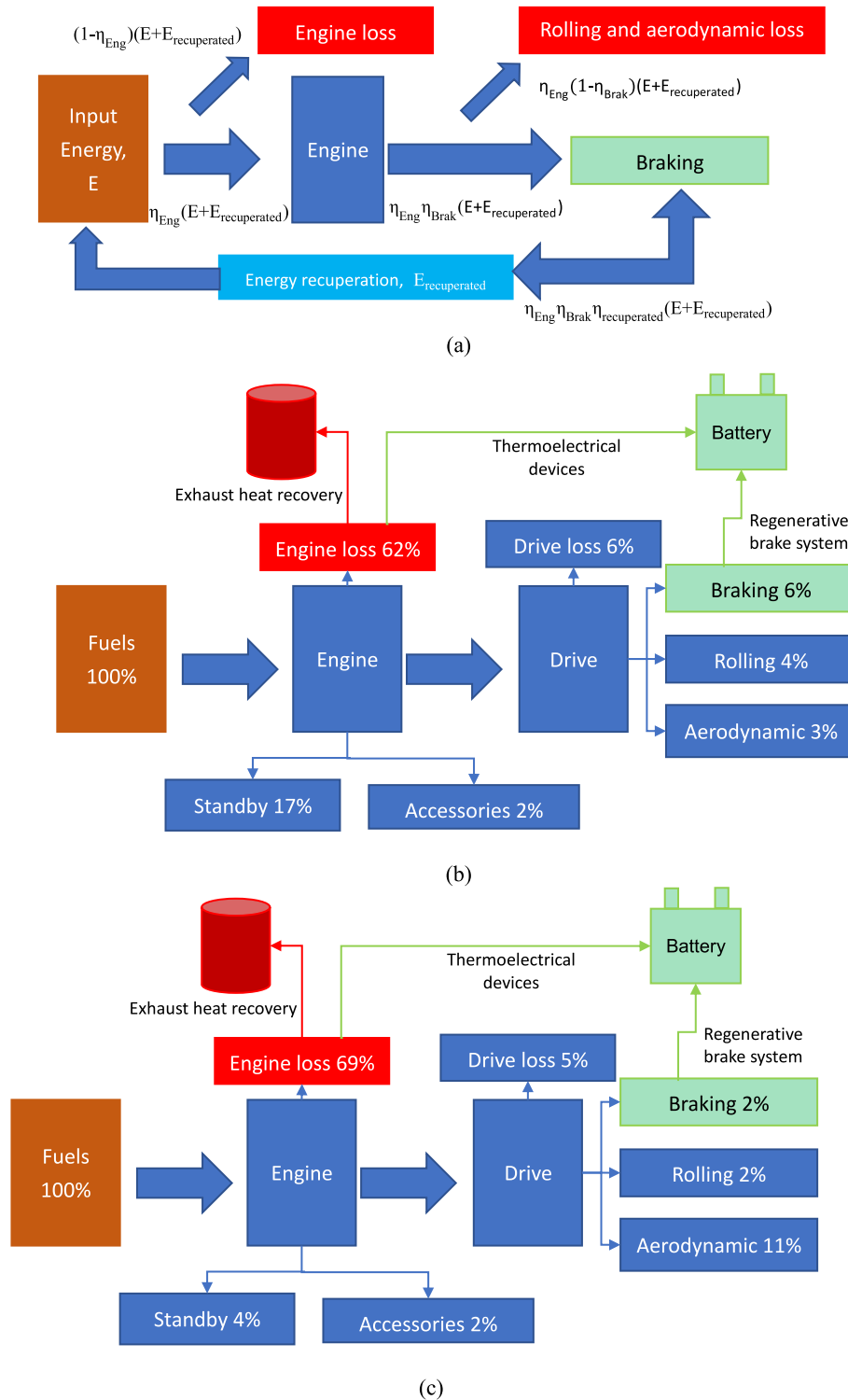


Fig. 8. Cascade energy efficiency of transportations: (a) a hierarchical topography for energy utilisation; (b) towns; (c) motorways [200,201]. (Note: η_{Eng} , η_{Brak} and $\eta_{recuperated}$ refer to the engine efficiency, brake efficiency, and recuperated efficiency).

3.5. Applications and future potential

District energy systems are reviewed on renewables, energy conversions and storages. Both DH and DC systems have been formulated and investigated, from perspectives of renewable-to-electric and thermal storages, energy management, and control strategy for multiple grids' interactions. Mobile vehicles, as mobile energy storages, can systematically connect different buildings, centralized energy storages,

exhaust heat sources (like factories), and energy generators (like CHP plants) [203]. Furthermore, grid-interactive buildings can promote load shed intensity and variability. Regarding the formulation of energy networks, the annular piping system is more reliable and flexible than the branched pipe network, in terms of power supply reliability, system resilience under extreme disruption, and mitigation on grids' pressure. However, several critical research challenges can be identified.

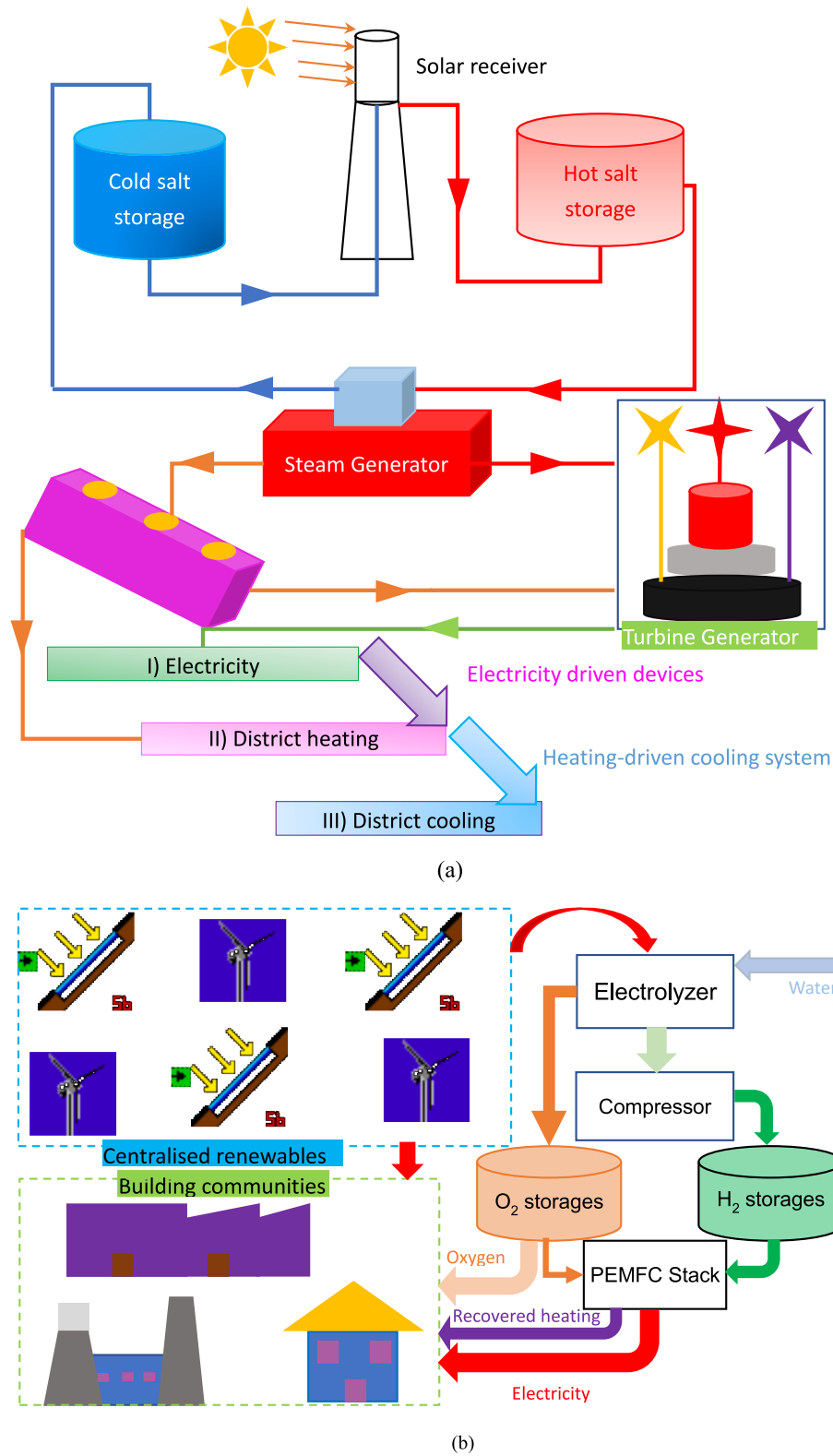


Fig. 9. (a) The diagram of the solar tower molten salt photothermal power generation system with hybrid grids' interactions; (b) the schematic diagram of renewable-fuel cell energy storage with hybrid grids' interactions.

- 1) peak power mitigation strategy when enlarging the energy boundary from a single building to building clusters;
- 2) synergies within hybrid grids to enhance resilience and flexibility of districting energy network;

- 3) hybrid grids' requirement in demand response and conflictions between different energy grids.

Future studies can focus on.

- 1) peak power mitigation strategy through combined cooling, heating and electrical energy storages, together with energy dispatch and power control strategies;
- 2) synergistic operation and compensation among different types of energy sources, in terms of energy density, and demand response time. Furthermore, transition from traditional centralised power system to distributed power typology will promote the energy resilience of the multi-energy system, when suffering from climate change or extreme events;
- 3) trade-off solutions among different stakeholders and grid utilities for cost-benefit allocations.

4. Smart district energy systems with optimal design, robust operation, energy controls and resilient grids' interactions

Optimisation methodologies of multi-energy systems in district levels were reviewed, demonstrated and discussed, to give an insight into theoretical mechanisms of metaheuristic optimizations. Afterward, trade-off strategies between computational complexity and optimisation accuracy are proposed to promote further development. Moreover, applications and future potential are clarified.

4.1. Optimizations of district energy systems for the system design and operation

In respect to diversified energy systems from the district level, energy systems are complex in design and operation due to multiple parameters and criteria. Generally, in-depth analyses of the sophisticated district energy systems can be classified as renewable and distributed integrations, system design and planning, and system operation and control. Research objectives are multi-dimensional, as listed in Table 6, including system efficiency [25], electric power saving [26], primary energy consumption [27], equivalent carbon dioxide emissions [27–29], economic considerations (including the investment cost [30] and operational cost [27]) and life-cycle analysis [28].

Table 6 lists a holistic overview of optimal design and robust operation of district energy systems in respect to multi-diversified energy systems. Parametric analysis and system optimisation were studied by researchers with respect to single and multi-objectives. In respect to the parametric analysis, researchers focused on practical solutions, such as optimal configurations of a power plant based on a decentralized heating system [27] and cloud-based demand-response strategies for district heating networks [30]. In respect to system optimisation, the adopted optimisation algorithms include linear programming [29], genetic algorithms [31], and hybrid mixed-integer linear with a three-stage heuristic [32]. The optimisation parameters include both design and operating variables. Manente et al. [26] optimized thermal energy storage to reduce the consumption of natural gas. The results indicate that, by charging the thermal storage with a waste incineration plant, the peak demand can be reduced, together with the decrease of natural gas consumption by 10 %. To achieve a resilient and efficient energy transmission network, Li et al. [176] systematically reviewed optimal design on distributed systems. Through the comprehensive literature review, most adopted optimisation algorithms for the optimal system design and reliable operation include mixed-integer linear program, multi-objective evolutionary algorithm, etc. However, due to the complexity of mathematical models through dynamic programming, the optimisation processes are time-consuming and internal memory-consuming, which puts a high requirement on both software and matched hardware, such as supercomputing equipments.

Note that many studies for renewable penetration in DH, transmission loss, and demand-side management have significantly simplified the load patterns in building simulation. To improve the optimisation performance with trade-off strategies between computational efficiency and accuracy, several researchers focused on the development of new optimisation engines, from the perspectives of

optimisation mechanisms. Heijde et al. [207] optimized a solar-driven DH energy system using a newly developed multi-objective optimisation toolbox, *modesto*. The computational efficiency can be improved by 10–30 times without sacrificing the accuracy of optimisation results. Ikeda et al. [208] proposed a new evolution-based constrained optimisation method to optimize district energy systems. Unlike traditional dynamic programming with the searching principle for all samples, the underlying mechanism for the performance enhancement is the stochastic move of individuals to other search points with uniform random numbers, following a classical genetic algorithm. Research results indicated that, the computational load of the proposed method is only 1/457 of the traditional dynamic programming, with the sacrifice of the accuracy only by 2.1 %.

4.2. Machine learning based technique in the resilient and smart district energy systems

Interdisciplinary studies between machine learning and district energy systems have attracted widespread attention in the academic world. Machine learning techniques can be used in the prediction of district energy demand [33] and robust model development [34,35], surrogate model development for uncertainty and optimisation analyses [36], optimal design on geometrical and operating parameters, efficient and robust energy management and control strategy [37]. Moreover, data-driven technique has also been applied in building energy retrofit. Ahmad et al. [209] predicted district energy demand from different levels, using machine-learning models. The machine-learning based model can provide prediction accuracy and adequate forecasting intervals in response to the requirement of smart grids. Wei et al. [33] systematically and comprehensively reviewed different data-driven based methods to predict building energy demand. The data-driven approaches are promising for the building energy study, including predictions on building demand and profile, district energy-consumption mapping, the benchmark of building clusters, retrofit strategies and guidelines. In respect to the robust model development, Sharifzadeh et al. [34] comparatively studied the robustness of different machine-learning methods, in terms of renewable energy prediction. The research results indicate that, compared to other models, the neural networks were more reliable for the prediction of electricity demand and renewable generations. AL-Musaylh et al. [35] investigated electric demand forecasting with an artificial neural network model. Research results indicate that, the hybrid ANN model generated a relative root-mean square error of around 3.85 %. In respect to a district energy network, Reynolds et al. [37] optimized the energy generation and demand management, by driving the artificial neural networks. Research results indicate that, through the optimisation, heat generation can be increased by 44.88 %, together with an increased profit of 8.04 %. Gallagher et al. [36] indicated that, the data-driven model can improve the accuracy for the uncertainty of energy-saving, with the reduced error at 51.09 %.

Fig. 10 demonstrates the evolution of papers using machine learning methods in energy-related research over the past ten years. The number of studies for machine learning applications in district energy systems has increased rapidly. As shown in Fig. 10(a) and (b), machine learning application for design and operation accounts for the largest share, whereas the proportion decreased from 52.1 % in 2010 to 40.4 % in 2019. The publication number on machine learning applications in performance prediction has increased from 106 to 1517, and the proportion from 6.0 % to 14.9 % over the last ten years. In addition, the machine-learning methods for the control strategy, energy management, uncertainty and sensitivity analysis have also increasingly attracted researchers' interest during the past ten years.

Furthermore, machine learning methods show promising prospects when integrated with optimisation algorithms for optimisation on nonlinear systems. The machine learning method can also be flexible to be integrated with newly developed optimisation algorithms. By flexibly

Table 6
A holistic overview of optimal design and robust operation of district energy systems.

	Types	Study	Research/ Optimisation problem	Energy system scale	Solution/Algorithm	Objective(s)	DHC type
Parametric analysis	Renewable&distributed integrations	Manente et al. [26]	Single-objective	District heating system	Thermal storage in a district heating network	Electric power saving potential	Geothermal and waste-to-energy district heating system
	System design and planning	Rosato et al. [27]	Multi-objective	A small-scale Italian district (6 typical Italian residential buildings and 3 schools)	Robust power plant configurations based on decentralized heating system	Energy, emission and cost	A centralized solar hybrid heating system with seasonal storage
	System operation and control	Vanhoudt et al. [30]	Single-objective	District heating networks with different energy storage configurations	Cloud-based demand-response strategies	Investment costs	District heating networks and the effect of different thermal energy storage
System optimisation	Renewable&distributed integrations	Kazagic et al. [204]	Multi-objective	District	Modular district heating	Economic and environmental performances	District heating system with CHP and renewable energy sources
		Talebi et al. [28]	Multi-objective	Existing and newly built communities [28]	Dynamic optimisation	Annual life cycle cost and CO ₂ emission	Hybrid community-district heating system
		Wang et al. [31]; Weinand et al. [32]	Single-objective	District	Genetic Algorithms [31] combinatorial optimisation approach (mixed-integer linear optimisation model and a three-stage heuristic) [32]	Annual net profit [31], minimum-cost [32]	Extraction-condensing CHP unit [31]; district heating networks [32]
	System design and planning	Tańczuk et al. [25]; Luo et al. [205]	Single-objective	Municipal district heating source [25]; district energy networks [205]	Techno-economical optimisation algorithm [25]; mixed integer linear programming [205]	Efficiency [25] and cost [205]	Gas turbine powered municipal district heating [25]; renewable supported district energy networks on a virtual island [205]
	System operation and control	Fazlollahi et al. [206]	Multi-objective	District energy systems	Mixed-integer linear programming	Environmental impact; total annual costs; penetration ratio of renewable energy [206]	Centralized and decentralized technologies
		Dorotić et al. [29]	Multi-objective	District	Linear programming	Discounted investment and operational costs; carbon dioxide emissions	District heating and cooling systems

integrating the machine learnt optimisation matrix in the teaching-learning-based optimisation algorithm, Zheng et al. [210] successfully identified the multivariable optimisation of an aerogel glazing system. According to the results, the total heat gain of an optimal system from the teaching-learning-based optimisation is 7.2 % lower than the total heat gain of an optimal solution through the particle swarm optimisation algorithm. The model predictive control can improve demand flexibility with great potentials in electricity cost saving and peak demand reduction.

4.3. Discussion on future prospects

Throughout the holistic and comprehensive literature review, the main concerns of machine learning techniques are on the optimal design and robust operation, control strategy and energy management, energy prediction, uncertainty and sensitivity analysis. Due to the intrinsic characteristics of computational efficiency with accurate feature extraction and classification, the upcoming avenues for future studies related to machine learning in district energy systems can be on.

- 1) surrogate model development to replace mathematical models without sacrificing the prediction accuracy;
- 2) short-term energy predictions for real-time energy control, and long-term prediction for model-predictive control;
- 3) generation of penalty signals from hybrid grids to improve techno-economic performances;
- 4) the fault detection and diagnosis of multi-energy systems in the entire energy chain, i.e., energy supply, transmission, transmission and distribution processes.

Several critical research challenges have been identified as avenues for future research. The identified challenges include the cost of a large amount of training data, the accuracy and robustness of developed machine learning model within advanced learning rules, synergic implementation of penalty signals from hybrid grids for the resilience and flexibility enhancement of districting energy network, hybrid grids' requirement in demand responsive and conflicts between different energy grids.

5. Outlook and recommendations

5.1. Technology implementations and practical applications

In the academia, researchers worldwide start to study machine learning-based digital district heating/cooling with renewable integrations and advanced low-carbon transition with preliminary research outcomes. For instance, Manfren et al. [211] studied data-driven building energy modelling with interpretability and generalisation. The approach can promote the scalability of data-driven techniques in district heating/cooling networks. For critical variables selection, Hwang et al. [212] proposed a systematic parameter selection approach based on domain knowledge and correlation analysis, promoting the prediction of electric energy consumption of heating and cooling systems. The cyber-physical systems and artificial intelligence can further promote the systemic resilience and energy security [213]. Meanwhile, research results in this study can also be widely applied in different climates. The cooling/heating network design, the demand and grid-responsive control, advanced energy conversions, and the flexible management can improve the energy resilience and flexibility, when

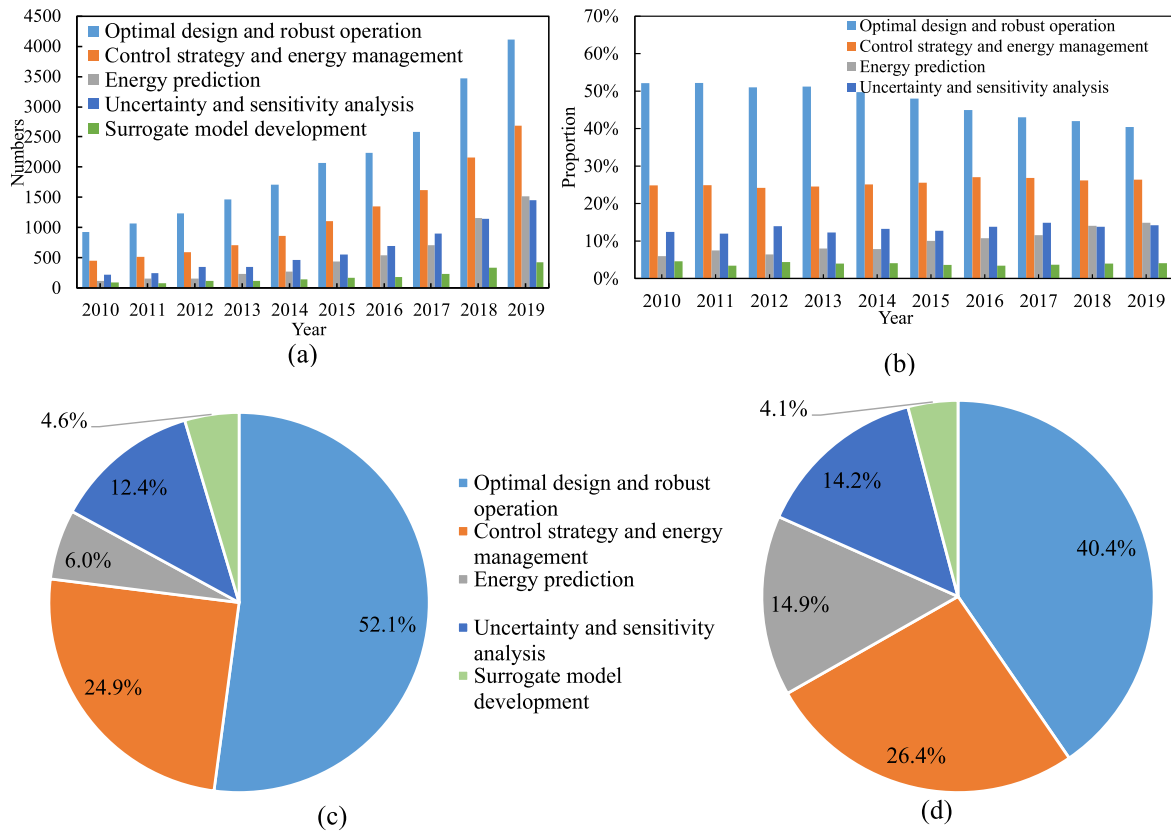


Fig. 10. Machine learning methods in district energy systems for different research purposes: (a) the number of research studies; (b) the proportion of research papers; (c) the constitution of studies in 2010; (d) the constitution of studies in 2019.

suffering from climate change or extreme events. Parametric analysis, single-objective and multi-objective optimizations can assist the optimal design and robust operation of district energy systems. Furthermore, machine learning technique can be used in the prediction of district energy demand and the robust model development, the surrogate model development for uncertainty and optimisation analyses, the geometrical and operating parameters' optimal design, and the efficient and robust energy management and control strategy.

The integration of various energy sources, storages (like plug-in EVs), heating/cooling/power/hydrogen grids makes the multi-energy system becomes complex in both structural configuration and accurate performance prediction. Challenging issues mainly include flexible interfaces for various components integration and temporal mismatch due to the simulation time inconsistency in convergence for each integrated

component. For example, it will take around three days to predict the one day thermodynamic and energy output performance of an aerogel-based PV/T-PCM system [214] (due to the iteration of sunlight and heat transmission within the nanoporous aerogel structure, heat transfer of latent phase change materials via finite volume method and enthalpy-porosity), while the end-users (like building owners) do not allow so long time for calculation as the end-users only care about the real-time supplied power. The machine learning can address this issue through the multiple linear regression, the support vector regression and the backpropagation neural network. The overfitting is solved by eliminating the unnecessary connections with small weights regularization, adding the sum of the weights to the learning function. Generally, the machine learning based techniques can play significant roles in future integrated multi-energy systems due to high computational efficiency,

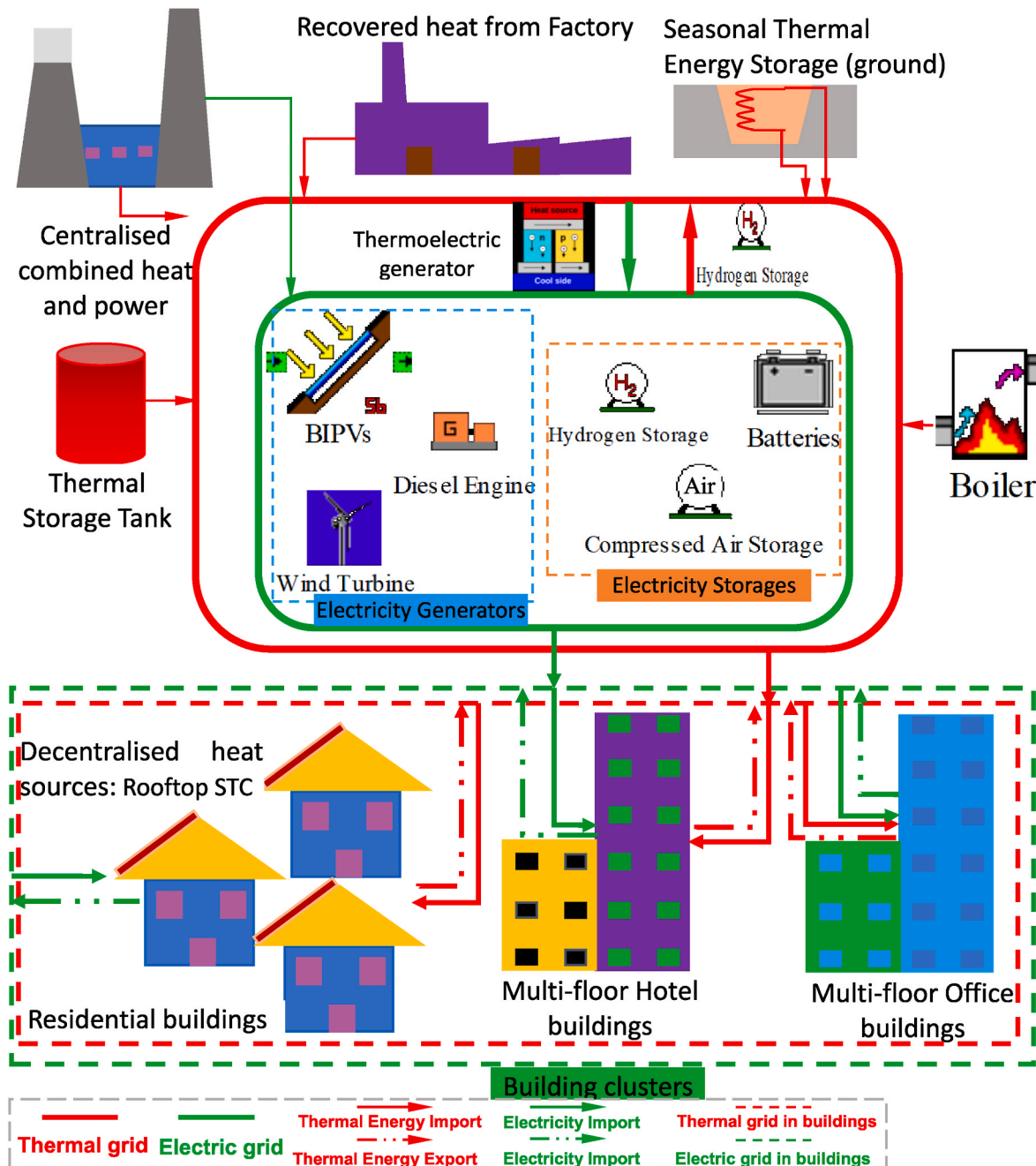


Fig. 11. A district energy system with renewable systems, diversified energy forms and hybrid energy storages.. (Note: the STC and BIPVs refer to solar thermal collector and building integrated photovoltaics.)

accurate performance prediction of nonlinear systems, user-friendly interface, including accurate performance prediction [215,216], advanced model predictive controls [217], single- and multi-objective optimisations [218,219].

5.2. Research limitations

Several research limitations of this study are summarized below.

- 1) advanced controls or strategies are seldomly applied in real systems, due to the user-unfriendly, heavy computational resources, mismatch between the model iteration and real-time reaction requirement of the system. For example, in the case when the computational time-duration for iteration until the convergence of model predictive controls is longer than the permitted system response time, the MPC cannot be effectively applied in cooling/heating systems;
- 2) enhancement strategies for energy resilience and flexibility of district heating and cooling networks have not considered the distributed energy system typology, smart interactions and effectiveness assessment;
- 3) nexus among energy digitalisation, energy network typology, infrastructures and big data have not been clearly provided.

Furthermore, along with the planning, design and operation processes of integrated multi-energy systems, error sources are multi-diversified, like uncertainties from renewables, stochastic energy-use behaviours and vehicle driving behaviours, electricity prices, energy policies, and so on. These factors will affect the predicted technical performances, optimal results, and economic-environmental performances. Policy makers may need to initiate incentives to promote the infrastructure installation, widespread adoption of advanced science technologies. Furthermore, the advancement in machine learning can improve the performance prediction accuracy in high efficiency, through multiple linear regression, support vector regression and backpropagation neural network.

5.3. Challenges and future studies

Throughout the systematic and comprehensive literature review, district energy systems with interactive energy sharing mechanisms show promising prospects for renewable integration with cleaner power production, mitigating grids' dependence and global warming, and the resilience improvement of smart energy systems. Both district heating and cooling networks have been formulated, integrating energy storages and conversions for the cascade energy utilisation. Renewable sources include solar photovoltaics, solar thermal energy, geothermal energy, biomass and recovered waste energy. The role of coal, diesel and natural gas can thereafter be transited from main sources to the backup. Within the district energy system, thermal energy storages include sensible, latent and thermochemical energy storages. Electric storages include battery storage, compressed air energy storage, supercapacity, fly-wheel and hydrogen energy storage. Fig. 11 demonstrates a general working principle of district energy systems with renewable systems, diversified energy forms and hybrid energy storages. Centralised (combined heat and power) and decentralized energy sources (solar thermal collectors in buildings, building integrated photovoltaics and diesel engines) are included to support the district energy system. Waste heat recovery from the centralized factor accounts for a large share of all energy sources, promoting the renewable and sustainable development of smart energy systems. Multi-diversified energy storage systems include hydrogen, compressed air storage and battery for real-time and daily electricity storage, ground for seasonal thermal storage purposes. Energy conversion units, such as a thermoelectric generator for thermal-to-electric conversion and a heat pump for electric-to-thermal conversion, are installed for cascade energy sharing between different energy grids.

From the perspective of the end-user, different types of buildings can be participated in the district energy system, including residential buildings, multi-floor hotel buildings and multi-floor office buildings.

In respect to the formulated district energy system, several challenges were presented for future studies.

- 1) Benchmarks to select the most suitable energy storage systems and advanced energy management strategies. Considering the mechanism difference of diversified energy storages and local conditions (such as climate and geographical conditions), appropriate energy storage systems can guide designers and stakeholders.
- 2) The conflicts of hybrid grids need to be well addressed, considering the local climates, users' needs and occupants' requirements. The synergic functions between different grids with diversified energy conversions can enhance the resilience and flexibility of hybrid grids for integrating intermittent renewable sources and covering stochastic demands.
- 3) The formulated district energy system involves multi-dimensional variables (such as the size of energy source and storage systems, the occupants' behavior, operating schedule of devices and so on), and multi-objectives (such as the energy efficiency, the initial and operational costs, the equivalent CO₂ emission, life cycle cost and emission, and so on). It is necessary to conduct optimisations to promote the development of renewable and sustainable district energy networks with high penetration of renewable energy.
- 4) The further update of advanced optimisation algorithms is critical to enhancing the overall performance of district energy systems.
- 5) The multi-criteria decision-making is quite complicated for all stakeholders participating in the district energy systems, including the end-users, building owners, facility managers, aggregators, district and transmission operators. Developing effective tools to help stakeholders make the 'best of the best' decision, along the Pareto optimal front is necessary, to improve their participation willingness and readiness.
- 6) Environmental, Science and Government (ESG) policies to cultivate associated technologies and promote its widespread applications in low-carbon district heating/cooling.

6. Conclusions

In this study, district energy systems have been systematically and comprehensively presented, in respect to district heating/cooling networks, hybrid renewables' integration, energy storages (such as thermal and electric storages) with heating, cooling and electrical energy forms and advanced energy conversions (such as the solar-to-heat, the solar-to-electricity, the heating-to-cooling conversions), and hybrid grids' interactions (such as district cooling/heating grids, power grid, natural gas grid, biofuel grid and hydrogen grid). An in-depth dialectical analysis on advantages and disadvantages, together with the techno-economic feasibility, has been provided, in respect to diversified thermal and electrical storages. Energy management and control strategy have been reviewed from perspectives of the design on district energy networks, the demand and grid-responsive control, and the flexible management of hybrid energy storages, in accordance with hybrid grids' requirements. Roles of hybrid renewables, novel energy storage systems, and diversified grids' interactions have been analysed. The main function of diversified grids' interactions is to realise the dynamic balance between hybrid renewables and energy demands and improve the grids' resilience to fluctuations and intermittence of renewable energy supply. Both single-objective and multi-objective optimizations of district energy systems have been comprehensively presented to promote smart district energy systems with optimal design, robust operation, energy controls and resilient grids' interactions. Furthermore, due to intrinsic characteristics of computational efficiency with accurate feature extraction and classification, machine learning methods show promising prospects for the realisation towards resilient and smart district energy systems, in

terms of prediction of district energy demands and robust model development, surrogate model development for uncertainty and optimisation analyses, optimal design on geometrical and operating parameters, efficient and robust energy management and control strategy. Due to the computational complexity of multivariable and multi-objective optimisations of district systems, machine learning methods can also be used to develop surrogate models to significantly reduce the computational load without sacrificing the accuracy of optimisation results. Last but not the least, a resilient and smart district energy network has been proposed to guide the design in the future city, involving on-site renewable generations, waste heat recovery from centralised power plants, multi-diversified energy storages, advanced energy conversions for the interactive energy sharing. Critical research results are summarized below.

- 1) In academia, energy storages in district energy systems include thermal energy storages (sensible, latent and thermochemical storages), and electricity storages (electrochemical battery, compressed air energy storage, supercapacitor, fly-wheel and hydrogen). Compared to latent or sensible heat storages, superiorities of thermochemical storages include higher energy density, higher operating temperature and higher efficiency. However, due to the high operating temperature, technical challenges include sophisticated system control, service life, material compatibility issues and chemical reactions. In respect to electricity storage, technical challenges include performance depreciation over long-time operations, operational security, enhancement of energy storage density, and energy loss during charging process. Furthermore, energy consumption from transportations needs to be considered when formulating smart district energy systems, in which the regenerative brake system is one of reliable and promising energy sources when integrating thermal and battery storages. The mobility of vehicles can improve regional renewable energy balance by overcoming uneven spatiotemporal distribution through smart charging/discharging technologies and energy sharing.
- 2) In order to promote resilient and smart district energy systems, district heating and cooling networks with diversified grids' interactions have been formulated. Technical challenges include the diversity of occupants' behaviours within different types of buildings, the consideration of hybrid grids' requirements and conflictions between different grids, and the peak power mitigation when enlarging the energy boundary from single building to a large scale, such as building cluster levels.
- 3) In order to improve the resilience and flexibility of district heating and cooling networks, technical solutions can be focused on the heating networks' design, the demand and grid-responsive control, and the flexible management of hybrid energy storages in accordance with hybrid grids' requirements. Furthermore, the activations of synergistic functions, considering differences of hybrid grids' requirement with diversified energy conversions, are effective solutions for enhancing the resilience and flexibility of multi-energy systems. From the perspective of heating networks' design, the district heating network with annular pipes is more reliable and flexible than the branched pipe network, in terms of the power supply and the mitigation on grids' pressure.
- 4) The main function of diversified grids' interactions is to dynamically balance the hybrid renewables and energy demands. Furthermore, the grids' resilience to fluctuations in the energy supply can be improved through advanced energy conversions between different energy grids in district energy systems. In respect to energy management, both demand-side management and energy storages can be operated to manage the renewable energy, in according to climate conditions, users' needs and grids' requirements. The availability of advanced energy conversions can promote the renewable penetration, energy sharing and grid' independence, in respect to diversified energy forms.
- 5) Parametric analysis, single-objective and multi-objective optimizations of district energy systems for the optimal design and robust operation have been investigated in the academia. The assessment criteria include system efficiency, electric power saving, primary energy consumption, equivalent CO₂ emissions, economic considerations (such as investment cost and operational cost) and life-cycle analysis. The adopted optimisation algorithms include the linear programming, the genetic algorithms, and the hybrid mixed-integer linear with a three-stage heuristic. However, due to the computational complexity and time-consuming of traditional optimisation methodology with the searching principle for all samples, the development of new optimisation engines to reduce the computational load without sacrificing the accuracy has attracted researchers' interest.
- 6) The interdisciplinary between the machine learning and district energy systems shows promising prospects for the promotions of resilient and smart district energy systems. The machine learning technique can be used in the prediction of district energy demand and the robust model development, the surrogate model development for uncertainty and optimisation analyses, the geometrical and operating parameters' optimal design, and the efficient and robust energy management and control strategy. Furthermore, in respect to the computational complexity and the time-consuming of optimisations, trade-off strategies between computational complexity and accuracy can be reached, through the operation of machine learning approach with optimisation algorithms.
- 7) In respect to the formulated district energy system with high energy flexibility and resilience, identified technical challenges include benchmarks for the selection of most suitable energy storage systems considering the mechanism difference of diversified energy storages and local conditions (such as climate and geographical conditions), conflictions of hybrid grids, optimisations with advanced optimisation algorithms, and multi-criteria decision-making to promote the willingness and readiness for stakeholders' participations.

The literature review and rational analysis show that renewables' integration, advanced thermal and electric energy storages, energy management and control strategy, and diversified grids' interactions, are critical elements of the resilient and smart district energy systems. Machine learning methods are promising. Future work can focus on the enhancement of resilience and flexibility of multi-energy systems with activations of synergistic functions of hybrid grids with diversified energy conversions, the generation of penalty signals from hybrid grids for the improvement of techno-economic performances, the flexible transition from physically and commercially isolated district energy systems to interactive energy sharing networks, and the machine-learning based optimisation methodology flexibly integrating advanced optimisation algorithms. This review can contribute to the regulation/standards to policy makers, applications, and policy targets in industry, countries/regions, considering energy, emissions/environment, costs, climate change, Environmental, Science and Government (ESG) and Sustainable Development Goals (SDGs).

CRediT authorship contribution statement

Yuekuan Zhou: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Siqian Zheng:** Validation, Visualization, Writing – original draft, Writing – review & editing. **Jan L.M. Hensen:** Validation, Visualization, Revision.

Declaration of competing interest

The authors declared that they have no conflicts of interest to this work. We declare that we do not have any commercial or associative

interest that represents a conflict of interest in connection with the work submitted.

Data availability

The data that has been used is confidential.

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